

TIANMING HUO

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Education

National University of Singapore

Aug 2020 – Jan 2025

Ph.D. in Industrial System Engineering and Management
with NUS Research Scholarship

Advisor: Cheung Wang Chi

Research Area: Operations Research, Revenue Management, Online Algorithms, Competitive Analysis, Resource Allocation, Order Scheduling, Dynamic Pricing

National University of Singapore

Aug 2015 – June 2019

B.Eng. in Industrial System Engineering and Management
Graduated with Honours (Distinction)

Research Papers

Optimal Online Allocation of One-Way Substitutable Resources

2026

Tianming Huo, Guodong Lyu, Wang Chi Cheung
under revision for *Operations Research*
Available on SSRN

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7109699

Online Reusable Resource Allocations with Multi-class Arrivals

2022

Tianming Huo, Wang Chi Cheung
Minor revision for *Operations Research*

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4320423

Online Lead Time Quotation with Predictions

2024

Tianming Huo, Wang Chi Cheung
forthcoming in *INFORMS Journal of Computing*

<https://pubsonline.informs.org/doi/10.1287/ijoc.2025.1104>

Optimal Markdown Pricing with Patient Customers

2025

Wang Chi Cheung, Tianming Huo

Major revision (2st round) for Production and Operations Management (POM)

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5522978

Online Reusable Resource Assortment Planning with Customer-Dependent Usage Durations

2023

Tianming Huo, Wang Chi Cheung

Available on SSRN

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4504713

Principal Component Analysis for High Dimension Stochastic Gaussian Process Model Fitting

2019

Maxime Xuereb, Tianming Huo and Szu Hui Ng

Invited Presentations

Online Reusable Resource Allocations with Multi-class Arrivals

- 2022 INFORMS Revenue Management and Pricing (RMP) Section Conference June, 2022
Virtual
- 2022 Analytics for X Conference Sept, 2022
Singapore
- 2023 POMS-HK International Conference Jan, 2023
Virtual
- 2025 INFORMS International Conference (Advance booking) Jul, 2025
Singapore
- 2026 POMS-HK International Conference (Advance booking) Jan, 2026
Hong Kong
- 2026 INFORMS Annual Meeting Nov, 2026
San Francisco, United States

Online Reusable Resource Assortment Planning with Customer-Dependent Usage Durations

- 2023 INFORMS Manufacturing and Service Operations Management (MSOM) Conference June, 2023
Montreal, Canada
- 2023 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2023
London, United Kingdom
- 2023 INFORMS Annual Meeting Oct, 2023
Phoenix, Arizona, United States
- 2024 POMS-HK International Conference Jan, 2024
Hong Kong

Online Lead Time Quotation with Predictions

- 2024 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2024
Los Angeles, United States

Optimal Markdown Pricing with Patient Customers

- 2024 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2024
Los Angeles, United States

Optimal Online Allocation of One-Way Substitutable Resources

- 2026 INFORMS Annual Meeting Nov, 2026
San Francisco, United States

Teaching

Teaching Assistant, National University of Singapore

- IE2100: Probability Models with Applications AY20/21, AY21/22, AY22/23 Sem 2
Teaching tutorial sessions, Grading homework, Holding office hours
- IE3101: Statistics for Engineering Applications AY21/22 Sem 1
Teaching tutorial sessions, Grading homework, Holding office hours
- IE5123: Reliability Engineering AY22/23 Sem 1
Grading Homework, Holding Office hours
- IE5121: Quality Planning and Management AY23/24 Sem 1
Teaching tutorial sessions, Grading Homework, Holding Office hours

Reviewing Service

Reviewer for *Management Science*, *Transportation Science*

Working Experience

NRF (National Research Fund) Research Fellow May 2026 – Present
Singapore Management University

- Lead research projects for the topic of Algorithmic Resilience in Smart City Operations: Real-time Resource Allocations for Urban Sustainability. Design real-time algorithms that incorporate ML-learned predictions to enable intelligent decision-making that is both data-informed and resilient against sudden disruptions.

Research Fellow
National University of Singapore April 2025 – May 2026

- Lead a research project to optimize the real-time allocation of resources under realistic settings, including designing algorithms, deriving theoretical results and conducting numerical experiments.

Capacity Industrial Engineer Aug 2019 – July 2020
Micron Semiconductor Asia Pte Ltd

- Achieved history high weekly backend throughput by improving workstation production rate and ensuring sufficient capacity at the Encapsulation/Post-Encapsulation segment of the line
- Identified capacity bottleneck and procured new workstation to improve overall capacity
- Coordinated workstation relocation project while ensuring no production impact
- Trained new team member on capacity IE calculation and model operation
- Rectified, updated and maintained capacity model to reflect precise data on production line

System Design Project Team Member Aug 2018 – Apr 2019
Softbank Robotics Singapore

- Increased footfall to shops experiencing low volume of customers through the design and deployment of a humanoid robot programme

Central Procurement Intern Jan 2018 – Jun 2018
Rolls-Royce Singapore Pte Ltd

- Developed Excel VBA macro tools to automate and optimize data analysis and reporting

Skills

Software Skills: Proficient in Python, Matlab. Familiar with C/C++, R
Language: Chinese (Native), English (Fluent)